

Quiz 10

Friday, December 4 — eigenvalues, complex numbers, and Lab 5 (Week 13)

Last updated: October 5, 2026.

Covers the prior instructional week only: [Eigenvectors](#), [Characteristic](#), [Eigenbehavior](#), and [Lab 5](#) (Week 13). Fall break is skipped when choosing the prior week. Same-day [SVD](#) / [Lab 6](#) / least squares (Week 15) are **not** on this quiz.

Textbooks: [Strang §6.1](#) · [Lachniet Octave PDF](#) · also on [Resources](#).

[Schedule](#)

SVD and least squares are not on this quiz; see [Lab 6](#), the [capstone packet](#), and the [final](#).

Core

- S 6.1 #1, #2, #5, #8, #10, #12, #14, #16, #17
- L Ch. 5 Ex. 1: selected hand/software eigenvalue parts
- L Ch. 5 Ex. 2: Markov equilibrium/eigenvector portions
- L Ch. 5 Ex. 5: selected matrix-power portion
- C10.1: Verify $AV \approx VD$ from software output.
- C10.2: Interpret a dominant eigenvalue and eigenvector in context.
- C10.3: Verify a proposed real or complex eigenpair by substitution.

Stretch

S 6.1 #3 and #4.

What a quiz item might change

A new 2×2 matrix with a factorable characteristic polynomial, a proposed eigenpair to verify, a Markov transition probability, or a rotation matrix with complex eigenvalues.